

AI/ML intern DAY 3

Task 2 Body Measurement Logic with Human Pose Detection

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Date: 13-05-2026

1. Shoulder Width Calculation

Introduction

Shoulder width calculation is an important part of AI-based body measurement systems used in fashion technology, tailoring, virtual try-on systems, and fitness applications.

AI systems estimate shoulder width using computer vision and pose detection technologies. Instead of manually measuring the body using measuring tape, AI can calculate approximate measurements from images or live video streams.

How Shoulder Width is Calculated

Pose detection systems such as MediaPipe identify body landmarks including:

- left shoulder
- right shoulder

These landmarks contain coordinate positions on the human body.

The distance between the left and right shoulder points is used to estimate shoulder width.

AI systems use:

- pose estimation
- distance calculation
- image scaling techniques

to calculate body proportions accurately.

Technologies Used

Common technologies used:

- OpenCV
 - MediaPipe
 - TensorFlow
 - Computer Vision
 - Deep Learning
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Applications

Shoulder width estimation is used in:

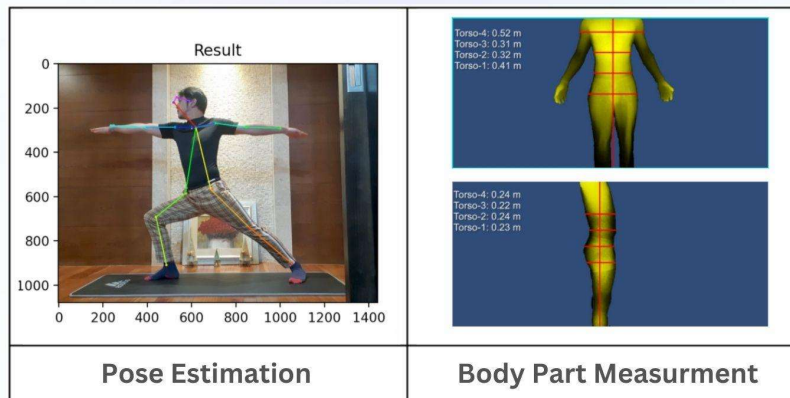
- clothing size recommendation
 - tailoring systems
 - virtual fitting rooms
 - fashion e-commerce
 - fitness tracking
-

Advantages

Benefits include:

- automatic body measurement
- faster size prediction
- reduced manual effort
- improved online shopping experience

Computer Vision Technology



2. Height Estimation

Introduction

Height estimation is the process of calculating a person's approximate height using AI and computer vision systems.

Traditional height measurement requires physical tools, but AI-based systems estimate height digitally using body landmark positions.

How Height Estimation Works

Pose detection systems detect body landmarks such as:

- head
- shoulders
- hips
- knees
- ankles

The AI system calculates the vertical distance between the top and bottom body landmarks.

Scaling algorithms and body proportion models are used to estimate approximate height.

Technologies Used

Height estimation systems commonly use:

- pose detection
 - image processing
 - machine learning
 - body scaling algorithms
 - computer vision
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Applications

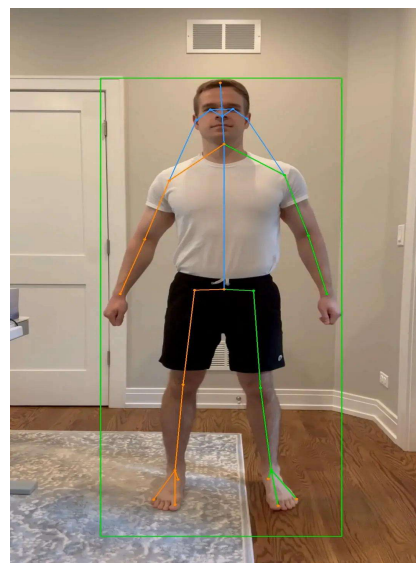
Height estimation is used in:

- virtual try-on systems
 - biometric identification
 - healthcare systems
 - fitness applications
 - smart surveillance systems
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Advantages

Benefits include:

- contactless measurement
- real-time analysis
- fast estimation
- automated body analysis



3. Waist Estimation

Introduction

Waist estimation is an AI-based technique used to estimate waist dimensions using pose detection and body contour analysis.

This technology is important in fashion, tailoring, healthcare, and fitness applications.

How Waist Estimation Works

The AI system analyzes:

- hip landmarks
- waist region
- body contours
- body proportions

Pose detection systems first identify body landmarks, then computer vision algorithms estimate waist dimensions using body shape analysis.

Some advanced systems also use:

- depth estimation
 - segmentation models
 - 3D body reconstruction
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Technologies Used

Technologies commonly used:

- MediaPipe
 - OpenCV
 - Deep Learning
 - Body Segmentation
 - Computer Vision
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Applications

Waist estimation is used in:

- online clothing stores
- health monitoring
- fitness tracking
- virtual try-on systems
- tailoring applications

Advantages

Benefits include:

- accurate clothing recommendations
- automated body analysis
- reduced manual measurements
- personalized shopping experience

